

Activity 10 Body mass index

Aim: Create a two-way table of values.

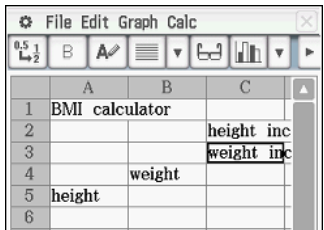
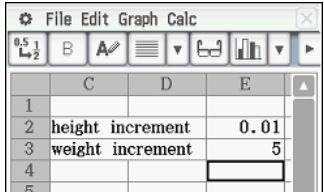
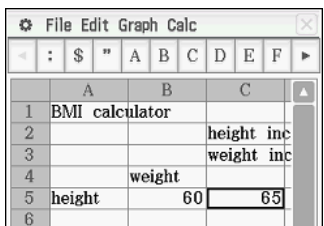
The Body mass index (BMI) is sometimes used as a health indicator.

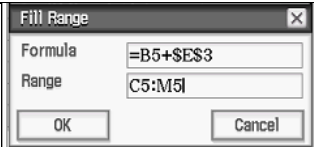
It is calculated by dividing a person's weight (in kg) by the square of their height in metres.

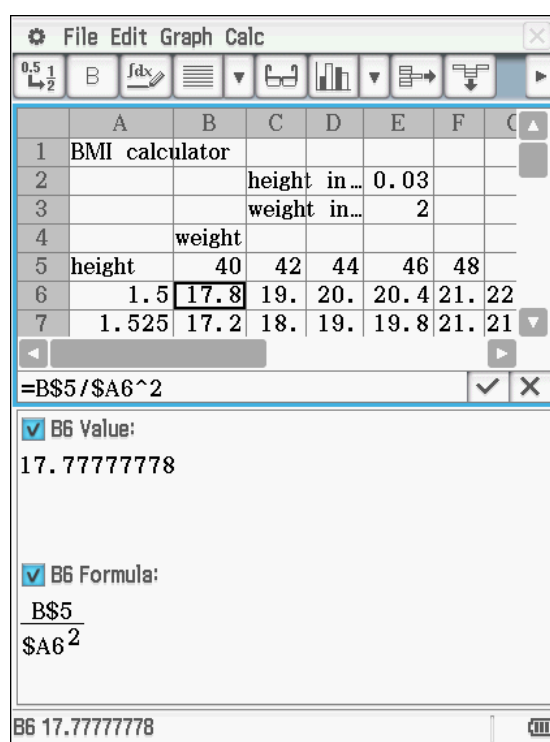
$$\text{BMI} = \frac{\text{weight}}{\text{height}^2}$$

1. Estimate your own BMI.
2. What is regarded as a healthy BMI for Australians of your gender?
Refer to this or other sources: <http://www.heartfoundation.org.au/healthy-eating/Pages/bmi-calculator.aspx>

Build a spreadsheet to display a two-way table or chart for looking up BMI values.

Enter headings • Tap in cell A1 and enter <i>BMI calculator</i> • Tap in cell C2 and enter <i>height increment</i> • Tap in cell C3 and enter <i>weight increment</i> • Tap in cell B4 and enter <i>weight</i> • Tap in cell A5 and enter <i>height</i>.	
Set increments • Tap in cell E2 and enter 0.01 This is the difference in height between successive rows • Tap in cell E3 and enter 5 This is the difference in weight between successive columns	
Enter weight column headings • Tap in cell B5 and enter 60 or whatever you want for the smallest weight. • Tap in cell C6 and enter the formula =B5+\$E\$3.	

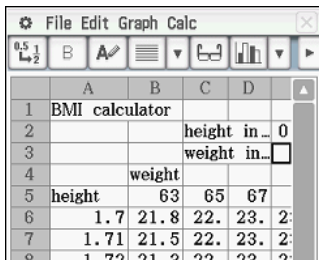
(using the menu bar is the quickest way to make the entry) • Tap in cell C6 • Select [Edit Fill Fill Range] • Set the range to C5:M5 • Tap OK	
Enter height row labels • Tap in cell A6 and enter 1.4, or whatever you want for the smallest height • Tap in cell A7 and enter the formula =A6+\$E\$2. • Tap in cell A7 • Select [Edit Fill Fill Range] • Set the range to A7:A20 • Tap OK	

Fill table • Tap in cell B6 and enter the formula =B\$5/(\$A6^2) • Tap in cell B6 • Select [Edit Fill Fill Range] • Set the range to B6:M20 • Tap OK	
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3. Set up your spreadsheet to display values so you are able to fill in the grid.

Adjust spreadsheet values

- Set the increment in E2 to 0.01
- Set the increment in E3 to 2
- Change the start height in A6 to 1.7
- Change the start weight in B5 to 63



Height	Weight (kg)			
	63	65	67	69
1.70				
1.71				
1.72				
1.75				

4. Complete the table

Height	Weight	BMI
1.78 m	75 kg	
164 cm	62 kg	
1.66 m		18.5
	85 kg	25